D2.1 Cell and nuclear division

NAME	DATE	TOTAL
		_____ / 48

About 50 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **STATE** the name given to the two DNA molecules held together at a centromere. [D2.1.5] [1]

2 **STATE** the proteins around which DNA wraps during condensation. [D2.1.6] [1]

3 **LIST** the four phases of mitosis, in order. [D2.1.7] [1]

4 **DEFINE** haploid and diploid. [D2.1.9] [2]

5 **STATE** two examples of unequal cytokinesis. [D2.1.3] [2]

6 **STATE** the number of chromosomes in a nucleus of a person with Down syndrome. [D2.1.10] [1]

7 **STATE** the phase in which homologous chromosomes form pairs. [D2.1.11] [1]

Section B: Understanding

17 MARKS

8 **EXPLAIN** why nuclear division occurs before cell division. [D2.1.4] [2]

9 **DISTINGUISH** between cytokinesis in animal cells and in plant cells. [D2.1.2] [3]

10 **EXPLAIN** how mitosis produces two genetically identical nuclei. [D2.1.5, D2.1.7] [4]

11 **EXPLAIN** why meiosis is needed in a sexual life cycle. [D2.1.9] [2]

12 **OUTLINE** the two rounds of segregation in meiosis. [D2.1.9] [3]

13 **EXPLAIN** how crossing over generates genetic variation. [D2.1.11] [3]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

14 [D2.1.7, D2.1.8]

A student counted onion root tip cells in three regions, at increasing distance from the tip.

REGION	INTERPHASE	PROPHASE	METAPHASE	ANAPHASE	TELOPHASE
A: CLOSEST TO TIP	160	22	8	4	6
B: FURTHER FROM TIP	185	9	3	1	2
C: FURTHEST FROM TIP	198	2	0	0	0

a CALCULATE the mitotic index for region A. [2]

b DESCRIBE the trend in dividing cells with distance from the tip. [1]

c SUGGEST why most cells in every region are in interphase. [1]

d DESCRIBE how you would identify a cell in anaphase. [2]

15 [D2.1.10]

The table shows live births with trisomy 21 by age of the mother.

AGE OF MOTHER / YEARS	25	30	35	40	45
BIRTHS WITH TRISOMY 21 PER 10 000	8	10	27	95	300

a CALCULATE how many times greater the rate is at 45 years than at 25 years. [1]

b DESCRIBE the relationship. [2]

c EXPLAIN how an error in meiosis results in trisomy 21. [3]

d A species has a haploid number of 5. CALCULATE the combinations possible from random orientation alone. [2]

Section D: Longer answer

8 MARKS

16 **COMPARE AND CONTRAST** mitosis and meiosis, including both what they share and what they produce. [8]

[D2.1.4–11]